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A CNN-transformer framework for emotion recognition in code-mixed English–Hindi data

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Abstract

Social media is an open platform for users to express their views and thoughts through emotions about a particular topic in natural language, leading to the generation of a vast amount of emotional data on micro-blogging sites. This data needs to be processed to extract meaningful insights and analyze emotions from text, enhancing various Natural Language Processing (NLP) applications. While humans can easily infer emotions due to their commonsense knowledge, machines lack this perception and must be trained to understand and detect emotions, making it a highly researched task in NLP. The challenge intensifies in Indian contexts where users often switch between two or more languages (code-mixing) to express opinions. The lack of annotated datasets for such multilingual data makes this a promising and underexplored area of research. To address this, we propose a hybrid deep learning framework using a CNN-Transformer model trained on Hindi–English code-mixed tweets, categorized into Happy, Sad, Anger, and Neutral emotions. Unlike previous approaches that rely solely on monolingual models or pre-trained transformers, our method combines local feature extraction via CNNs with global contextual modeling through Transformers specifically designed for code-mixed structures. We also utilize pre-trained word embedding fine-tuned on the dataset to improve semantic representation. Our proposed model achieves superior performance, with an F1-score of 0.82 and outperforming both CNN-only and Transformer-only baselines, and demonstrates robust emotion classification in code-mixed social media text.

Keywords Deep learning, Natural language processing, Emotion, Code-mixed, Embeddings

1 Introduction

Language serves as a primary medium of communication among human beings, enabling the expression of opinions, reviews, and emotions in a natural and intuitive manner. The use of one's native language fosters greater comfort and authenticity during such interactions. Social media platforms, in particular, have emerged as dynamic spaces where individuals from diverse backgrounds articulate their thoughts, emotions, and opinions through text, gestures, and speech. This has resulted in the generation of an enormous volume of emotionally rich data, particularly through micro-blogging sites



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such as Twitter, where users frequently communicate in natural language. The analysis and interpretation of this emotional data have become increasingly significant. Emotion detection from textual data enables a deeper understanding of public sentiment, trending topics, and behavioral patterns. Such insights have numerous applications, including enhancing human–machine interaction, supporting mental health initiatives through early detection of concerning emotional patterns, enabling sentiment analysis for strategic decision-making related to customer satisfaction, and facilitating personalized interventions in e-learning environments by monitoring students' emotional states.

While humans possess innate common sense and can detect emotions almost instantaneously, machines lack the perceptual and world knowledge required for such intuitive understanding. Consequently, training computational models to recognize and predict human emotions has emerged as a critical research objective within the domain of Natural Language Processing (NLP). Among various approaches, the application of Machine Learning (ML) and Deep Learning (DL) techniques has shown considerable promise in bridging the gap between human communication and machine comprehension. In recent years, Artificial Intelligence (AI) has witnessed significant advancements, gaining widespread popularity due to its ability to perform complex, human-like tasks. Deep Learning, a prominent subset of AI, has demonstrated substantial potential across multiple technological fields, including software development, data analysis, and autonomous systems. Previous research efforts predominantly focused on monolingual datasets, largely owing to their greater availability. However, as communication patterns evolve, an increasing number of users now incorporate code-mixed language in their social media interactions. This phenomenon is particularly prevalent in multilingual countries such as India, where linguistic diversity is deeply rooted in cultural practices. Code-mixing, the embedding of words, phrases, or sentences from one language into the utterances of another, has become a natural characteristic of online communication.

In India, Hindi is the most widely spoken language, and its combination with English is commonly observed on social media platforms. This study addresses the complexities associated with detecting emotions in Hindi–English code-mixed texts, a task that remains relatively unexplored due to the initial scarcity of suitable datasets. To illustrate the challenges associated with code-mixed text analysis, we present two examples extracted from a Hindi–English code-mixed dataset sourced from Hindi newspaper social media posts:

- Text 1: "Hume bhi team me lelo, England ghoomna hai"
Translated: "Take us in your team, we want to visit England."
- Text 2: "Bhai woh injured hai test cricket nahin khel sakta"
Translated: "Brother, he is injured and cannot play test cricket."

Text 1 conveys excitement and happiness, while Text 2 provides factual information indicative of a neutral emotional tone. These examples highlight the semantic richness and syntactic variations inherent in code-mixed texts, which present substantial challenges for automated emotion detection.

The primary objective of this research is to detect and classify emotions in code-mixed English–Hindi textual data using deep learning algorithms and to benchmark the performance against state-of-the-art approaches. For this purpose, we employ a manually annotated code-mixed dataset, categorizing the entries into four emotional classes:

happy, sad, anger, and neutral. Understanding emotions in this context facilitates deeper public sentiment analysis and enables the identification of early warning signals related to social unrest, mental health concerns, and political instability. To accomplish this, we propose a Convolutional Neural Network (CNN)-Transformer hybrid model and evaluate its performance in comparison to standalone CNN and Transformer models.

The structure of the paper is as follows: Sect. 2 presents the literature review; Sect. 3 outlines the summarized findings; Sect. 4 describes the dataset and preprocessing steps; Sect. 5 details the proposed methodology; Sect. 6 presents the experimental results; and Sect. 7 concludes the study with suggestions for future research directions.

2 Literature review

In pursuit of effective emotion detection using Deep Learning (DL) techniques, several studies have explored various Machine Learning (ML) and DL algorithms. Kadir et al. [1] designed a hybrid sentiment analysis model by integrating ML algorithms with a pre-trained Bidirectional Encoder Representations from Transformers (BERT) model for the Turkish language. They utilized Term Frequency-Inverse Document Frequency (TF-IDF) and word vectors, achieving a sentiment analysis accuracy of 91.05%. Amirreza et al. [2] conducted a comprehensive study on DL algorithms such as Boltzmann machines, Deep Belief Networks, and autoencoders, highlighting their challenges and applications in Natural Language Processing (NLP). Similarly, Daniel et al. [3] compared three Recurrent Neural Network (RNN) models—Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (BiLSTM), and Gated Recurrent Unit (GRU)—for mood and emotion detection tasks, finding the GRU model to be a particularly strong performer. Onur et al. [4] investigated content-based Short Message Service (SMS) classification for the Turkish language using both ML and DL methods, achieving an accuracy of 99.86% with a Convolutional Neural Network (CNN) model. Joshua et al. [5] analyzed the prediction of film box office revenues and reviews using NLP and DL techniques, reporting satisfactory performance.

Dong-Min et al. [6] explored emotion detection within movie scripts using an LSTM model, evaluating the model's performance through the calculation of average cosine similarity between predicted and target sentiment values. Daniel et al. [7] provided an extensive analysis of various DL architectures for NLP tasks across multiple application domains. Miryam et al. [8] studied communication patterns among English- and Spanish-speaking Twitter users, applying traditional ML algorithms alongside DL models such as Extreme Gradient Boosting (XGBoost) and CNNs for mental health condition identification, achieving notable results. Chen et al. [9] presented a comprehensive review of advancements in Large Language Models (LLMs) for human-agent interactions, highlighting emerging research directions and applications across domestic and industrial sectors.

Vankayala et al. [10] focused on depression detection using social media texts, leveraging a hybrid DL model that combined word embeddings, LSTM, and CNN architectures to achieve robust performance. Mattia et al. [11] proposed a fusion of semantic, DL, and NLP techniques for enabling natural language interaction between humans and humanoid robots, specifically in the development of the autonomous robot "Zora." Adithya [12] compared three bilingual word embedding approaches for English–Hindi code-mixed datasets, emphasizing the need for multilingual embeddings to support further research.

Tulasi et al. [13] conducted emotion detection on Hinglish datasets using CNN, LSTM, and BiLSTM models. Similarly, Satyajit et al. [14] addressed hate speech detection within English–Hindi code-mixed datasets by comparing three different DL models. Deepanshu et al. [15] contributed to corpus creation and annotation for emotion detection in code-mixed datasets, while Aditya et al. [16] employed Support Vector Machine (SVM) and Random Forest algorithms for processing code-mixed language data. Lastly, Samira et al. [17] explored the application of Transformer models for multimodal emotion recognition tasks, further expanding the scope of emotion analysis across diverse input modalities.

The reviewed studies collectively underscore the growing significance of DL techniques in emotion detection tasks. However, the majority of prior research has predominantly focused on monolingual datasets. Limited attention has been directed toward the challenges inherent in code-mixed language settings, particularly in the context of Hindi–English data. This gap presents a compelling opportunity for further investigation, particularly through the development of models capable of effectively processing and analyzing code-mixed emotional data.

3 Summarized findings

The review of related work highlights a growing demand for the development of bilingual and multilingual embeddings to support effective processing of code-mixed datasets. Word embeddings have demonstrated their utility as powerful feature representations across numerous NLP tasks. With the increasing prevalence of code-mixed language use on social media platforms, particularly in linguistically diverse regions such as India, the need to analyze such data effectively for emotion detection has become increasingly evident. Recent advancements in DL models—including Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), Transformers, Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM)—have significantly transformed the landscape of textual data processing. These models are capable of learning complex linguistic patterns and have proven to be effective for emotion recognition in large-scale datasets.

The studies reviewed consistently underscore that the integration of DL techniques with word embeddings and NLP preprocessing methods yields promising results. In particular, hybrid architectures combining multiple DL models have shown enhanced performance in sentiment and emotion classification tasks. Given the linguistic complexity and semantic richness of code-mixed data, especially in the Hindi–English context, such integrated approaches offer a promising path forward. Building upon these insights, the current research proposes the use of a CNN-Transformer hybrid model for the task of emotion detection in English–Hindi code-mixed texts. Word vectors are employed as input features, and the model is trained to classify emotions into four categories: happy, sad, anger, and neutral. This classification enables more granular analysis of public sentiment and emotional trends in multilingual social media discourse.

4 Dataset description

Initial work on English Hindi code mixed dataset was done by Vijay et al. [15] and we consider using the same data set. We also extracted the sentences/tweets with emojis and without emojis using Twitter API related to the category of emotions we are

interested in, for identifying various feature traits to train our model and considered classifying emotions into four classes namely happy, sad, anger and neutral for our study. (<https://pypi.python.org/pypi/twitterscraper/0.2.7>). We extracted total 20,781 tweets related to the four emotions and the tweets are uniformly distributed for classification. Tweets retrieved from social media are in csv format and consist of all kind of information like emoticons, special characters, mentions, likes, ids, punctuations, URL etc. We received tweets featuring various emotions and they were written either in English language or Hindi or code mixed. We removed all the tweets which were purely written in English or Hindi language and retained code mixed tweets. As the study is focused on emotion detection, we considered only those code-mixed tweets which were expressing emotions related to the category of hatred, happy, sad and neutral. Total text extracted along with dataset from Vijay et al. [15] accumulated to 20,102 texts. Table 1 shows detailed description of code-mixed data set. There is a need to preprocess these tweets for further processing.

4.1 Preprocessing

Cleaned words are converted into tokens or subwords and preprocessed tweets are made ready for feature extraction. Following steps were performed for preprocessing.

1. **Removal of Twitter handle(@user):** Twitter handles are removed for a cleaner input
2. **Removal of URLs:** As the URLs don't significantly contribute towards detecting of emotions, they are removed from the text.
3. **Removal of emoticons:** As the study is focused on detecting emotions from text, we removed all the emoticons and are not used in feature extraction.
4. **Removal of Hashtag:** We observed many hashtag like #cricket were removed.
5. **Retaining text case:** There were many words which were written in upper case and sentence case which sometimes carries emotions weights in tweets. Hence we retained the text case.
6. **Normalized the text:** We removed all punctuations, extra spaces and repeated characters to ensure smooth feature extraction further.
7. **Keep Code mixed structure intact:** As the code mixed dataset contains emotional clues and we wanted to retain the realistic input pattern, we retained the original code mixed dataset.

5 Proposed methodology

To ensure clarity, reliability, and reproducibility in the tweet emotion annotation process, we adopted a systematic and transparent approach: The tweets were annotated with the following four emotion categories:

- **Happy:** Representing emotions like excitement, joy, and celebration.
- **Sad:** Representing feelings of disappointment, sadness, or regret.

Table 1 Dataset distribution

Classes	Tweets
Happy	5533
Sad	2091
Anger	683
Neutral	11,795

- **Anger:** Representing frustration, criticism, or sarcasm.
- **Neutral:** Representing factual, objective, or non-emotional content.

5.1 Annotation process

We employed a **semi-automatic annotation approach** using a pretrained transformer-based emotion classification model:

- **Model Used:** cardiffnlp/twitter-roberta-base-emotion (available via HuggingFace).
- The model outputted fine-grained emotion labels, such as joy, sadness, and anger.
- These were **mapped** to our simplified 4-category system based on semantic relevance and emotional intensity.

5.2 Quality control

To ensure reliable and consistent labelling:

- **Manual Validation:** 20% of tweets were randomly selected and manually reviewed by two independent annotators.
- **Inter-Annotator Agreement:** The agreement was measured using **Cohen's Kappa**, which resulted in a score of **0.78**, indicating substantial agreement.
- Any disagreements were resolved through discussion, ensuring consensus.

To capture the semantic information of both Hindi and English words in code-mixed text, we employ a word embedding approach that maps individual tokens to dense vector representations. Specifically, we utilize Continuous Bag of Words (CBOW) embeddings as described in Kamble et al. [14], which are trained on code-mixed Twitter data. CBOW is chosen due to its ability to effectively capture contextual similarity, where semantically related words are mapped to similar regions in the vector space. This property is particularly beneficial for handling code-mixed data, as it helps bridge lexical gaps between Hindi and English, and allows the model to learn meaningful bilingual patterns. Dense word embeddings also enhance the feature space by preserving linguistic structures and reducing sparsity, thus facilitating better learning during model training. These words are converted into vectors. The model is retrained using pre trained bilingual embedding to generate a more enhanced word vector. The retrained model generates word vectors that are enriched by leveraging prior linguistic knowledge embedded in the bilingual embedding. These enhanced embedding capture semantic and syntactic nuances across two languages, ensuring that similar concepts are closer in the vector space regardless of language.

6 Working

Figure 1 illustrates the workflow of the proposed emotion detection model for text. The pipeline begins with preprocessing of raw text data, including tokenization, normalization, in code-mixed content. From the cleaned text, feature vectors are extracted, which are then passed into a hybrid deep learning architecture for classification. The core of our approach is a hybrid deep learning model that integrates Convolutional Neural Networks (CNNs) and Transformer-based architectures. This combination is designed to leverage the strengths of both local and global feature extraction methods. CNNs are particularly effective at capturing local textual patterns such as emotional n-grams, especially in code-mixed texts (e.g., Hindi–English). These patterns can include specific

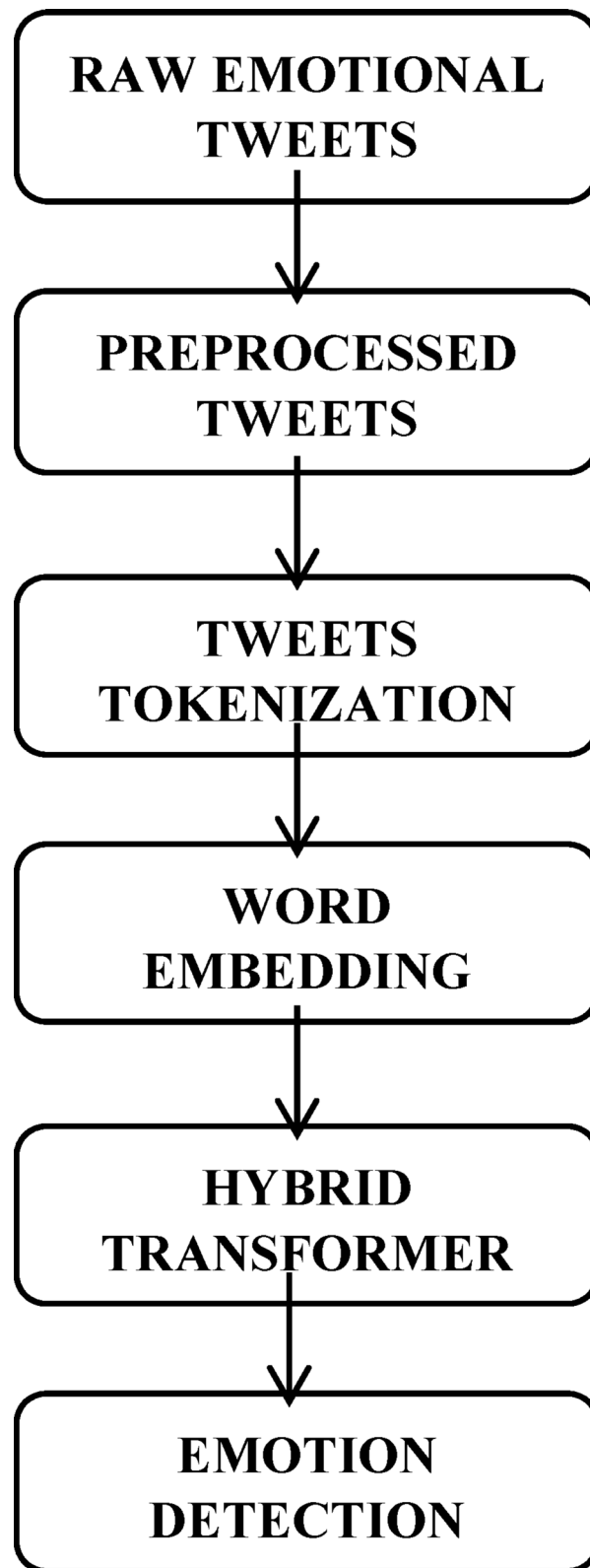


Fig. 1 Emotion detection pipeline

emotional phrases or bilingual expressions that are indicative of sentiment or emotion. The convolutional layers act as filters that highlight local dependencies and patterns that may be lost in models relying solely on sequential or attention-based approaches. Transformers, on the other hand, excel at modeling long-range dependencies and contextual information. They use self-attention mechanisms to understand the semantics of a sentence in a more holistic manner, which is critical in code-mixed language where emotion may be expressed across languages and span distant tokens in the text. Transformers help resolve cross-lingual ambiguities and provide a richer understanding of the context.

By integrating CNNs and Transformers in a hybrid architecture, our model benefits from both fine-grained local analysis and broader semantic comprehension. This is particularly advantageous for identifying complex emotional expressions such as sarcasm, mixed sentiments, or context-dependent emotional cues that are common in informal, multilingual communication. The model is trained using a labeled dataset of code-mixed text annotated with four emotion classes: happy, sad, angry, and neutral. To address the significant class imbalance present in the dataset—particularly the underrepresentation of *sad* and *angry* emotions—we applied class weighting in the loss function. Specifically, we used a weighted cross-entropy loss, where each class's weight was inversely proportional to its frequency in the training set. This strategy ensured that the model placed greater emphasis on correctly classifying minority emotions, thereby improving its overall robustness. The performance of the model is evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. This dual-layered approach allows for robust classification performance, outperforming baseline models that use either CNN or Transformer alone. The performance of the model is evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. This dual-layered approach allows for robust classification performance, outperforming baseline models that use either CNN or Transformer alone.

While several previous studies have explored emotion detection in monolingual or bilingual contexts using traditional or deep learning approaches [3, 6, 10, 11], relatively few have tackled the specific challenges posed by code-mixed data, especially in emotion classification tasks. Early works like Tulasi et al. [13] and Deepanshu et al. [15] utilized traditional classification methods on handcrafted features or basic word embeddings but did not leverage deep hybrid architectures that can effectively model both local and contextual dependencies. More recent approaches such as Adithya et al. [12] focused on generating embeddings for code-mixed languages but stopped short of integrating advanced architectures for emotion detection. In contrast, our method presents a hybrid CNN-Transformer framework that explicitly addresses the structural complexities of code-mixed Hindi–English text. The CNN layers are tuned to capture local bilingual emotion expressions (e.g., code-switched n-grams), while the Transformer component handles long-range contextual cues—a combination particularly effective for informal multilingual communication. Moreover, unlike prior deep learning methods that either fine-tune pre-trained models like mBERT [17] or rely on monolithic architectures, our dual-branch approach is tailored for the code-mixed setting, integrating class-weighted training to handle imbalanced emotion categories. This design provides a more robust and scalable solution to emotion recognition in multilingual social media contexts.

Table 2 Performance metrics (CNN only model)

Emotions	Precision	Recall	F1 score
Happy	0.79	0.68	0.76
Sad	0.75	0.59	0.66
Anger	0.72	0.42	0.53
Neutral	0.86	0.95	0.90

Table 3 Performance metrics (Transformer only model)

Emotions	Precision	Recall	F1 score
Happy	0.84	0.78	0.81
Sad	0.80	0.72	0.76
Anger	0.76	0.60	0.67
Neutral	0.89	0.95	0.92

7 Experiments

The dataset was randomly split into 80% training and 20% testing subsets. All the collected tweets were tokenized and converted into 300-dimensional vectors using a pre-trained bilingual word embedding model namely fastText and fine-tuned on code mixed dataset. The embeddings were then fed into a hybrid CNN-Transformer model. In the CNN component, we experimented with 2, 3, and 4-g convolutional filters to capture short-range emotion expressions. Each filter size was assigned 4 filters, totaling 12 filters per tweet. We used a fixed sequence length of 10 tokens. The CNN output was then passed to a Transformer encoder with 2–3 self-attention layers to model long-range contextual dependencies and better interpret complex tweet structures. This combination allowed the model to understand both local n-gram patterns and global sentence semantics for effective emotion classification. Max pooling is done to retain the important cues from the pattern which helps in emotion detection. The outputs from both networks were concatenated and passed through a dense layer followed by a softmax classifier. The model was trained with a learning rate of 0.001, a batch size of 32, and early stopping was applied based on validation loss to prevent overfitting. Our study analyzes the performance of CNN only model on 20,201 tweets to see how efficiently CNN captures local features. Table 2 reports the performance metrics obtained using the CNN-only model for emotion classification. Class weights were computed and applied during training to mitigate the effects of class imbalance.

The limited data on certain emotions resulted in less recall especially anger detection being weakest due to low sample size. As the dataset is imbalanced, the model failed to capture local features towards other classes than neutral class. For Anger, which is represented by only 683 tweets, CNN struggle to identify sufficient emotional patterns, as the convolutional filters failed to learn useful patterns from such a small set.

Embeddings are applied to transformer only model to analyze the performance and Table 3 presents the performance metrics for Transformer only model. It is observed from the below mentioned table that dataset imbalance still affects the accuracy of the model. The self-attention mechanism of transformer model helps capture emotional cues and relies on deep contextual understanding which helps improves the accuracy of the model.

As the study considers hybrid DL model for emotion detection, we pass the output of CNN to the transformer for final results. Transformer model helps in identifying

Table 4 Performance metrics (Hybrid model)

Emotions	Precision	Recall	f1 score
Happy	0.87	0.82	0.84
Sad	0.83	0.78	0.80
Anger	0.77	0.67	0.71
Neutral	0.92	0.96	0.94

Table 5 Comparison metrics (All three model and Baseline)

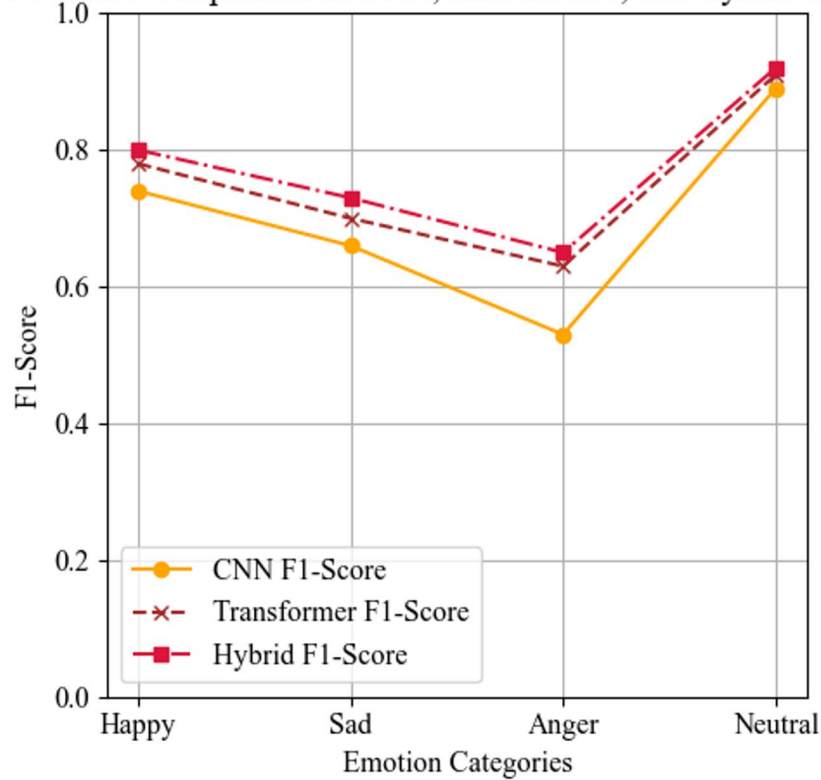
Models	Precision	Recall	F1 Score
CNN only	0.780 ± 0.004	0.660 ± 0.006	0.710 ± 0.005
Transformer only	0.820 ± 0.005	0.760 ± 0.004	0.790 ± 0.004
Hybrid approach	0.840 ± 0.006	0.800 ± 0.005	0.820 ± 0.005
mBERT (code mixed dataset)	0.830 ± 0.005	0.780 ± 0.006	0.801 ± 0.006

relationships between features using self-attention mechanism and produces single contextual embedding which is passed to the classification layer for detecting emotions. Code-mixed datasets often involve complex word-level structures and localized patterns due to multiple languages being interwoven in the same text. CNN helps capture these local patterns well and hence CNN was used prior. Code-mixed datasets require a deep understanding of long-range dependencies and semantic relationships. Transformers, with self-attention mechanisms, excel at capturing such relationships across an entire sequence, even in a multilingual setting. We trained the model for 17 epochs and the performance evaluation is done using precision, recall and F1 score and the performance metric is represented in Table 4 below.

In our comparative analysis, we evaluated the performance of three models for emotion classification using an imbalanced dataset of 20,201 code-mixed tweets labeled as Happy, Sad, Anger, or Neutral. All models were trained for 17 epochs with class weights applied to reduce bias toward the majority class. Our proposed Hybrid CNN-Transformer model outperformed both the CNN-only and Transformer-only baselines. To further assess its effectiveness, we also compared it with the mBERT model, which was fine-tuned under the same conditions on the same dataset. Although the hybrid model showed slightly better results than mBERT, we conducted additional analysis to confirm whether this difference was statistically meaningful. We ran each model five times using different random seeds and calculated the average performance along with 95% confidence intervals for each metric. We also performed a paired t-test on the F1-scores of the hybrid model and mBERT. The hybrid model achieved a mean F1-score of 0.820 ± 0.005 , compared to 0.801 ± 0.006 for mBERT, and the p-value from the t-test was below 0.05, indicating that the improvement is statistically significant. These results confirm that the hybrid model consistently performs better, especially for minority emotion classes like *Sad* and *Anger*, and that the performance gain is not due to random chance. We present a comparison Table 5 highlighting the average performance of all the three models.

We present the line graph below for our three models in Fig. 2 showing only the F1-Scores for CNN, Transformer, and Hybrid models across the emotion categories. It is clearly seen that hybrid model outperforms the other two models. The graph highlights the performance differences for each model, with the hybrid model generally showing better balance across all metrics, particularly in improving recall for *Sad* and *Anger*. The reason for misclassification is some tweets had overlapping semantics and resulted in ambiguity in understanding them. For example, the tweet “*Yakeen nahi ho*

F1-Score Comparison of CNN, Transformer, and Hybrid Models

**Fig. 2** F1 score**Table 6** Comparison with state of art models

Model	Precision	Recall	F1 score
CNN-LSTM [13]	0.8285	0.8281	0.8283
CNN-BILSTM [13]	0.8323	0.8321	0.8319
CNN [14]	0.8334	0.7851	0.8085
SVM [16]	0.7494	0.6315	0.684
RANDOM FOREST [16]	0.6243	0.5888	0.6060
CNN-Transformer (our framework)	0.84	0.80	0.82

raha, just wow" is ambiguous because *wow* represents happiness and *Yakeen nahi* may convey sadness. Also, tweets being incomplete, the context is not clear which leads to such ambiguity. Another reason for misclassification is multi label nature of some tweets. For example, the tweet "*Kab se 28 run pe hi tike hain*" is tagged as neutral but it also conveys some sadness which resulted in multi label nature and system is trained for one label classification. Some tweets lacked explicit emotional indicators, leading to confusion between *neutral* and *sad* classes. For example: "*Bas ho gaya ab*" (It's enough now)—depending on context, this could imply frustration, sadness, or neutrality. Code-mixed tweets with sarcastic tones were often misclassified. The model struggled to infer sarcasm without explicit cues. For example "*Wah bhai, kya baat hai!*" (Wow bro, that's something!)—often intended sarcastically, but classified as *happy*.

Table 6 provides a comparative analysis of the performance measures across various existing models and the proposed CNN-Transformer hybrid framework. It is evident that the proposed model achieves superior results with a precision of 0.84, recall of 0.80,

and F1 score of 0.82. While the results from CNN-LSTM [13] and CNN-BiLSTM [13] are very competitive and close to our framework, the marginal improvement achieved by our model can be attributed to the effective integration of CNN and Transformer components. CNN enables strong local feature extraction from short tweets, while the Transformer enhances the model's ability to capture contextual dependencies even in code-mixed data. Additionally, models such as SVM [16] and Random Forest [16] demonstrate lower performance, reflecting the limitations of traditional machine learning approaches in handling complex linguistic variations present in code-mixed text. Similarly, while CNN [14] performs well in precision, it shows comparatively lower recall, which affects the overall F1 score. Therefore, the proposed CNN-Transformer framework strikes a balanced trade-off between precision and recall, leading to an overall superior F1 score, which is critical for emotion detection tasks where both correct detection and coverage of emotional expressions are important.

8 Conclusions

Social media has become open platform for individuals express their emotions on various topics through text or images. In multilingual country like India, people often use a mix of English and their local language, commonly referred to as code-mixed language, to convey their emotions. Automatically identifying emotions in such code-mixed languages is challenging due to the distinct characteristics of the languages involved. In this study, we propose a DL approach to detect emotions expressed in Hindi-English code-mixed language on social media platforms like Twitter. For this purpose, we collected and pre-processed a dataset of 20,201 Hindi-English code-mixed sentences from various sources, annotated with four emotions: happy, sad, anger and neutral. To transform these sentences into vectors, we utilized a bilingual pre-trained model, which was retrained to generate more enhanced vectors. We evaluated the performance of the model by passing the vectors to the hybrid DL model, A CNN -Transformer framework was used for emotion detection, and the model demonstrated superior performance, achieving an accuracy of 85.31%. This model also showed robust performance in class-wise emotion detection. Our analysis highlights the importance of using a bilingual pre-trained model for processing code-mixed data. Tweets are often short and convey incomplete, ambiguous and sometimes overlapping semantic information, there is a need to remove the ambiguity from the tweets so that it becomes easy to detect emotions unambiguously. In the future this work can be extended to resolve ambiguity for Indian code-mixed languages. Handling ambiguity in tweets for emotion detection in code-mixed dataset is complex due to code mixing and transliteration, short and informal nature of tweets, spell correction, ambiguous context and lack of grammar rules. Code-mixed text inherently suffers from ambiguity at lexical, syntactic, and semantic levels, often due to polysemous words, inconsistent grammar, and informal usage. Resolving ambiguity for polysemy words can improve the accuracy to a greater extent.

Author contributions

SP wrote the main manuscript, MP prepared the figures and tables.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest

The authors declare no competing interests.

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